Question	Answer	Marks	Partial Marks
2(d)	0.5 or $\frac{1}{2}$	**3**	**M1** for $y = \frac{k}{(x+2)^2}$ oe **B1** for $k = 50$ or **M2** for $2(3 + 2)^2 = y(8 + 2)^2$ oe
2(e)	$\frac{7x-x^2}{2(x-2)}$ or $\frac{7x-x^2}{2x-4}$ oe final answer	**3**	**M1** for $5 \times 2 - (x - 5)(x - 2)$ oe seen **M1** for common denominator $2(x - 2)$ oe isw
3(a)	Rotation 90[°] clockwise oe Origin oe	**3**	**B1** for each
3(b)(i)	Image at (–4, –1) (–4, –4) (–2, –4)	**1**	
3(b)(ii)	Image at (3, –1) (5, –1) (3, –4)	**2**	**B1** for translation by $\begin{pmatrix} 7 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -5 \end{pmatrix}$ or for 3 correct points not joined
3(b)(iii)	Image at (–2, ½) (–2, 2) (–1, 2)	**3**	**B2** for 3 correct co-ordinates soi in working or correct size and orientation in wrong position or **M1** for $\begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}\begin{pmatrix} -4 & -4 & -2 \\ 1 & 4 & 4 \end{pmatrix}$ shown or for statement: enlargement, sf 0.5, (0, 0)
4(a)	$\frac{1}{2} \times 4(x - 1) \times (2x + 5)[\sin 90] = 30$ oe	**M1**	
	$8x^2 - 8x + 20x - 20$ or better	**B1**	correct expansion of brackets
	Completion to $2x^2 + 3x - 20 = 0$	**A1**	with no errors or omissions seen
4(b)	$(2x - 5)(x + 4)$	**M2**	Allow **M2** for e.g. $2x(x + 4) - 5(x + 4)$ then $2x - 5[= 0]$ and $x + 4[= 0]$ **M1** for $2x(x + 4) - 5(x + 4)$ or $x(2x - 5) + 4(2x - 5)$ or $(2x + a)(x + b)$ [= 0] where $ab = -20$ or $a + 2b = 3$ [a, b integers]
	2.5 and –4 cao	**B1**	

Question	Answer	Marks	Partial Marks
4(c)	11.7 or 11.66 … or 11.67	**3**	**M2dep** for $(4(their 2.5-1))^2+(2\times their 2.5+5)^2$ or **M1dep** for $4(their 2.5-1)$ or $2\times their 2.5+5$ OR **B1** for $\sqrt{20x^2-12x+41}$ and **M1dep** for substituting x = *their* 2.5 into $\sqrt{20x^2-12x+41}$ at any stage
5(a)	–3, 17	**2**	**B1** for each
5(b)	Fully correct curve	**4**	**B3 FT** for 10 or 11 points or **B2 FT** for 8 or 9 points or **B1 FT** for 6 or 7 points
5(c)(i)	Correct ruled tangent for *their* curve through (0, −17)	**1**	
5(c)(ii)	(1.7 to 2.2, –1 to 2.5)	**1**	
5(c)(iii)	[y =] $9x$ – 17 final answer	**3**	**M2dep** for answer [y =] $9x$[+] – c OR **M1dep** for gradient = $\frac{rise}{run}$ for *their* tangent at any point **B1** for answer [y =] kx[+] – 17 ($k \neq 0$)
5(d)	$y = 3x + 2$ ruled correctly and –2.2 … to –2.1 –0.6 to –0.4 2.6 to 2.8	**4**	**B2** for $y = 3x + 2$ ruled or **B1** for [y =] $3x + 2$ soi or $y = 3x + k$ ruled or $y = kx + 2$ but not $y = 2$ **B2** for all 3 values or **B1** for 2 values
6(a)	0.6	**1**	
6(b)	50.7	**3**	**M2** for $1.2 \times 19 + \frac{1}{2}(19 + 12) \times 1.8$ oe or **M1** for method for finding any relevant area
6(c)	17.9	**3**	**M2** for *their* 50.7 – 1.2 × 19 [– 10] oe or **M1** for 1.2 × 19 oe seen isw
7(a)	29	**1**	
7(b)	128	**2**	**FT** 180 – 2 (55 – *their* **(a)**) **M1** for angle *OCA* or angle *OAC* = 55 – *their* **(a)** soi